Sreejato (Sree) Chatterjee

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Education

University of Rochester Rochester, NY
B.S. in Computational Biology (Honors in Research), B.A. in Data Science *Aug 2022 – May 2026*
Honors Thesis: Epigenetic Regulation of Brain Aging GWAS Variants within Transposable Element Sequences

Publications

1. **Chatterjee, S.**, Tran, L., Nguyen, Q. D., Kirson, R., Hamlin, D., Aquino, H. "Assessing Historical Structural Oppression Worldwide via Rule-Guided Prompting of Large Language Models." *Proceedings of the 2025 IEEE International Conference on Big Data (BigData).* doi: 10.1109/BigData66926.2025.11401510.

Research Experience

National Library of Medicine (NLM) Bethesda, MD
Postbaccalaureate IRTA Research Fellow (Advisor: Dr. Richard Scheuermann) *August 2026 – Present*

- Incoming position; research focus in single-cell genomics and biomedical AI.

University of Rochester School of Medicine & Dentistry (SMD) Rochester, NY
Department of Biomedical Genetics (Advisor: Dr. Hongbo Liu) *May 2025 – May 2026*

- Developed an integrative genomics framework to characterize the regulatory landscape of noncoding brain-aging GWAS variants using RepeatMasker, single-nucleus ATAC-seq, and ENCODE cCRE annotations.
- Built scalable bioinformatics pipelines in Python and R integrating GWAS, chromatin accessibility, and transposable element datasets for enrichment analysis and pathway characterization.
- Applied partitioned LD Score Regression (S-LDSC) to evaluate heritability enrichment and demonstrate that TE-overlapping GWAS signals are largely explained by shared epigenetic regulatory architecture.

University of Rochester School of Medicine & Dentistry (SMD) Rochester, NY
Department of OB/GYN & Public Health Sciences (Advisor: Dr. Timothy Dye) *Sep 2024 – May 2026*

- Developed a rule-guided large language model framework to classify structural oppression from free-text ethnicity self-identification data across 170+ countries.
- Benchmarked instruction-tuned LLMs against expert-annotated labels across multiple prompting strategies to evaluate interpretability, validity, and cross-country generalizability.
- Applied a validated rule-guided LLM framework to over 11,000 multilingual survey responses from a global COVID-19 dataset to generate large-scale structural disadvantage scores for epidemiological analysis.
- Performed multivariable statistical analyses linking AI-derived social disadvantage measures to healthcare access, social determinants, and health outcomes.

National Institute of Neurological Disorders & Stroke (NINDS) Bethesda, MD
Summer Research Intern (Advisor: Dr. Mark Wagner) *Jun 2024 – Aug 2024*

- Investigated cerebellar learning mechanisms underlying dopamine self-stimulation behavior in transgenic mice models.
- Developed closed-loop computational pipelines for real-time behavioral tracking and optogenetic stimulation using Bonsai and Python.
- Applied DeepLabCut neural network models for pose estimation and behavioral quantification in conditioned place preference experiments.
- Designed algorithms to quantify reward-associated behavioral patterns and visualize locomotor trajectories across experimental learning periods.

National Cancer Institute (NCI) Bethesda, MD
Summer Research Intern (Advisors: Dr. Jeffrey Schlom, Dr. Caroline Jochems) *Jun 2023 – Aug 2023*

- Investigated combination immunotherapy strategies targeting HPV-related malignancies in preclinical murine models.
- Measured cytokine and chemokine expression within tumor microenvironments and peripheral tissues using

fluorescence assays and qPCR.
- Quantified tumor immune cell populations and evaluated anti-tumor immune responses across experimental treatment groups.

Oral Presentations

1. **Chatterjee S.**. "Multiomic Integration Reveals Regulatory Function of GWAS Variants within Transposable Elements in Brain Aging." In *International Conference on Intelligent Biology and Medicine (ICIBM)*, Buffalo, NY, 2026.
2. Dye, T., **Chatterjee, S.** "Measuring Global Group Disadvantage in Free-Text Identity Data." *Florida Atlantic University Summer Institute in Computational Social Science (SICSS)*, Boca Raton, FL, 2026.
3. **Chatterjee S.**. "Measuring Global Structural Oppression in Free-Text Identity Data via Rule-Guided Large Language Models." In *UR Undergraduate Research Expo Social Sciences Speakers Symposium*, Rochester, NY, 2026.
4. **Chatterjee S.**. "Measuring Global Structural Oppression in Free-Text Identity Data via Rule-Guided Large Language Models." In *oSTEM National Conference*, Baltimore, MD, 2025.

Selected Poster Presentations

1. **Chatterjee S.**. "Epigenetic Regulation of Brain Aging GWAS Variants within Transposable Element Sequences." In *National Conference on Undergraduate Research (NCUR)*, Richmond, VA, 2026.
2. **Chatterjee S.**. "Measuring Global Structural Oppression in Free-Text Identity Data via Rule-Guided Large Language Models." In *7th National Big Data Health Science Conference*, Columbia, SC, 2026.
3. **Chatterjee S.**. "Integrative Genomic Analysis Reveals Function of GWAS Variants Within Transposable Elements in Human Diseases." In *URMC Aging Research Day Conference*, Rochester, NY, 2025.
4. **Chatterjee S.**. "Measuring Global Structural Oppression in Free-Text Identity Data via Rule-Guided Large Language Models." In *oSTEM National Conference*, Baltimore, MD, 2025.

   Additional poster presentations at *Finger Lakes Science and Technology Showcase* (2025, 2026), *UR Undergraduate Research Poster Expo* (2024, 2026), *NIH Summer Poster Day* (2023, 2024), and *NCI Center for Cancer Research Poster Day* (2023).

Honors & Awards

- **Charles L. Newton Prize, University of Rochester Hajim School of Engineering (2026)**
  Awarded for exceptional undergraduate engineering research and scholarly achievement.
- **Third Place Winner, UR Undergraduate Research Lightning Talks Competition (2026)**
  Selected among 100 submissions and 20 finalists for a 3-minute research presentation competition.
- **Region Q Outstanding Student Service Award, Alpha Phi Omega (2026)**
  Recognized for exceptional service leadership and community engagement.
- **Research Presentation Award, UR Office of Undergraduate Research (2026)**
  Received $1,500 award supporting presentation of research at NCUR.
- **Medallion Leadership Society Inductee, University of Rochester (2025–2026)**
  Recognized for leadership development and campus inclusion programming.
- **First Place, UR Biomedical Data Science Hackathon, Undergraduate Division (Fall 2025)**
  Developed a gene expression prediction model for combinatorial perturbations.
- **Neurodiversity Alliance Scholarship Recipient (2025)**
  National scholarship supporting high-achieving neurodivergent students.
- **Second Place, Rutgers RAISE AI & Informatics Competition, Undergraduate Division (2025)**
  Led NLP analysis of 14,000+ news headlines; placed 2nd out of 119 teams.
- **Gold Award, Best Visualization Category, RIT ASA Datafest (2025)**
  Recognized for predictive analytics and data visualization on commercial real estate datasets.
- **First Place, UR GIDS Biomedical Data Science Hackathon, Undergraduate Division (Fall 2024)**
  Built a machine learning model for gene-regulatory associations.

- **Out 4 Undergrad Life Sciences Conference Scholar (2024)**
  Selected for a competitive national conference supporting LGBTQ+ students in life sciences.
- **Dean's Scholarship Recipient, University of Rochester (2022–2026)**
  Merit-based scholarship for academic excellence and leadership.

Technical Skills

- **Programming:** Python, R, Bash, SQL, Java
- **AI/ML:** PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Statsmodels
- **NLP & Text Mining:** Hugging Face Transformers, LLMs, spaCy, BERTopic, VADER, TextBlob
- **Bioinformatics:** Seurat, ScanPy, Bowtie2, PLINK, LDSC, BEDTools, RepeatMasker, GWAS, ENCODE
- **Imaging & Behavior:** DeepLabCut, Bonsai, ImageJ
- **Tools & Systems:** Git, Linux, Conda, Jupyter
- **Lab Techniques:** PCR, qPCR, Gel Electrophoresis, Western Blotting, Cell Culture, Flow Cytometry, DNA Extraction, Immunohistochemistry, Tissue Sectioning, ELISA, Fluorescence Microscopy

Leadership & Service

University of Rochester Out in STEM (oSTEM) — Rochester, NY
Co-President (2025–26), Secretary (2024–25) — *Jan 2024 – May 2026*

- Organized pre-professional programming and conference travel support for LGBTQ+ students in STEM.
- Coordinated graduate application workshops, faculty panels, and networking focused on research careers.

University of Rochester Alpha Phi Omega, Mu Lambda Chapter — Rochester, NY
Co-Vice-President of Service (2025–26), Secretary (2024–25) — *Jan 2024 – May 2026*

- Coordinated community service initiatives and nonprofit partnerships supporting regional outreach efforts.
- Organized workshops and large-scale service events for the 2026 Region Q Sectionals Conference.

University of Rochester Neurodiversity Alliance — Rochester, NY
Co-Chapter Leader (2023–26) — *Sep 2023 – May 2026*

- Led mentorship and advocacy programming supporting students with learning disabilities and neurodivergent learners.

University of Rochester Office of Residential Life & Housing — Rochester, NY
First-Year Fellow — *Sep 2023 – May 2025*

- Organized academic and wellness events, including faculty panels and study sessions for first-year students.

University of Rochester Chemistry Department — Rochester, NY
Teaching Assistant — *Jan 2024 – May 2024*

- Led workshops and office hours for CHEM 132 covering thermodynamics, kinetics, quantum mechanics, and problem-solving strategies.